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Harmony and algorithm

Exploring the advancements and impacts of AI-generated music

Tsen-Fang Lin and Liang-Bi Chen

Artificial intelligence (AI)-generated music has become a topic of great interest in the contemporary music industry. With the increasing development and popularization of AI technology, AI-generated music is no longer just a future prediction but a reality. The emergence of AI can help creators explore new possibilities during the creative process, find fresh sources of inspiration, and break free from the traditional framework of music structure. Additionally, AI-generated music provides an affordable solution for general music playback needs. Therefore, the changes in the music industry caused by AI investment are significant and warrant attention.

The source of AI music creation

The meaning of *composition* itself is to combine and organize the sound fragments in consciousness. Therefore, combining and organizing sound data are a kind of music creation, but who decides this process? Examples of handing over the mastery of music creation to the human brain have long existed: looking back on the history of human music creation, Mozart once composed a song, "Dice Music," in the classical period (Hedges, 1978) for which he automatically selected a combination of bars by throwing dice. The resulting music is still pleasant to listen to, but there is a certain randomness to the creative process. In addition, the "music of opportunity" promoted in academic circles in the 20th century heralded the possibility of AI creation in the future, just waiting for the arrival of machine learning (ML).

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The development of AI music dates back to 1948–1949, when Alan Turing created the first computer-generated musical notes in Manchester (Copeland and Long, 2017). By the 1980s, computer technology made music creation more convenient, and people began to study automatic music synthesis technology. In 1993, the first academic conference on music and AI, the International Conference on Artificial Intelligence in Education, was held in Edinburgh, Scotland, U.K. By 2004, IBM's "Computer Jazz Musician" produced the first piece of music automatically composed by AI. By 2023, many AI music generation platforms, such as Google's MusicLM, Meta's MusicGen, SoundDraw, and Jukebox, will have been opened to the market. AI music creation has now become a work partner (or competitor?) that even musicians have to face up to.

Sound database, popular music, and AI music creation

Digital music creation has long been how most people listen to music today. The comprehensive popularization of the sound databases required for digital music has paved the way for the future of generative music creation. When pop music becomes the mainstream of human auditory culture with the development of technology, business, and society, it is easy to understand that the resources for pop music production will become the focus of the equipment market. With the rise of electronic dance music, which is becoming strongly mainstream, the assembly and adjustment of samples and loops have become an essential method of music production. This type of creation, which has a stable speed and beat and completes musical content through combination, is particularly suitable for analysis, calculation, and reorganization. Because the most detailed beauty of music has been prepared in advance in the material, only essential selections and modifications are needed to turn it into a potentially pleasant creation.

We take the screens created by loop stacking in the famous BandLab software as an example (see Fig. 1). Straightforward block selection allows novices with no musical background to make a pleasant self-composed song when they start quickly. This does not mean that the critical selection and modification are easy tasks. Still, compared to all free creations starting from a single note and the sound of an instrument, modular creation is relatively more straightforward to "analyze." Therefore, after generative AI is opened to the public, it can produce more stable and higher quality music types that primarily work with a sample pack orientation. This type of music is dominated by electronic dance music. Therefore, it is unsurprising that current AI-generated music often performs well in the electronic dance music category.

Since most music can be divided into three aspects—melody, accompaniment, and rhythm—if the logic of creation is classified and organized, it is theoretically possible to use algorithms to create works close to real people. However, using one method alone may not lead to better results. For different needs, combining different calculation models may achieve better results. In addition, since more traditional music creation can be expressed in musical notation, it is pretty suitable to perform calculations by marking events, such as pitch and dynamics. With this way of calculating scores and rules to generate music, the famous country-type computer random

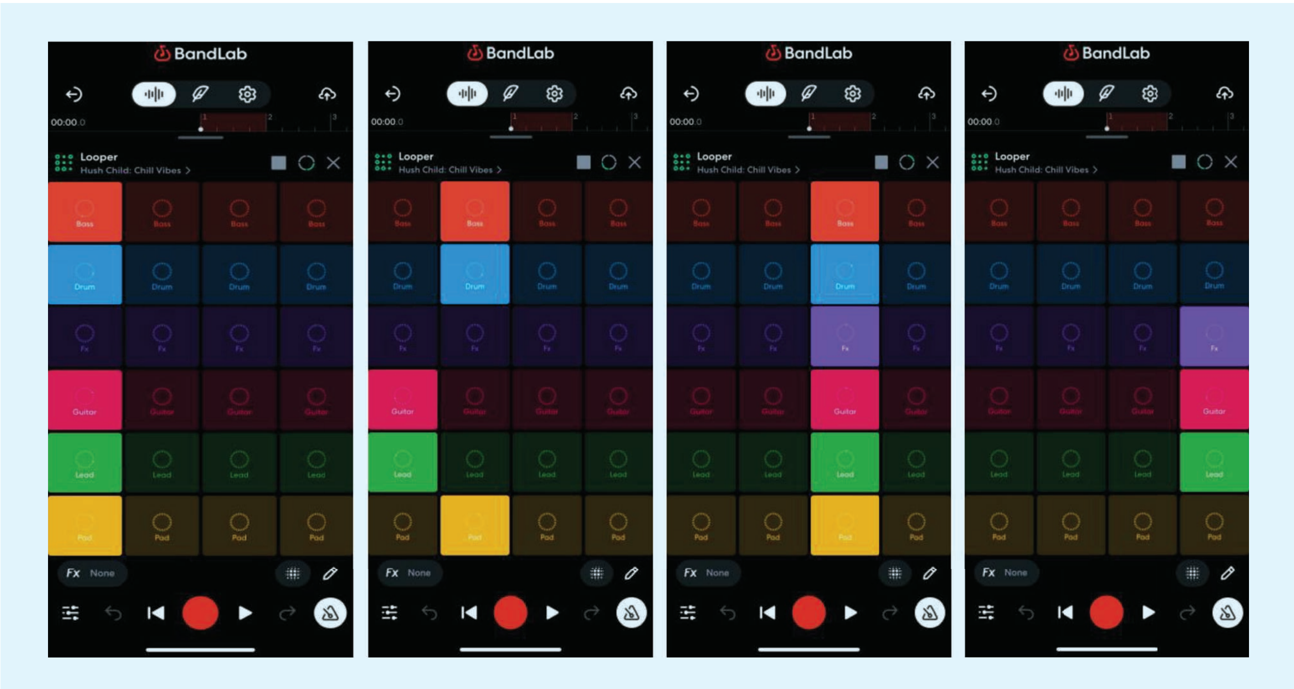

FIG1 "Looper" function screens of BandLab.

creation software Band-in-a-Box reached a very high level many years ago. Introduced in 1990 by PG Music Inc. (https://www.pgmusic.com/), Band-in-a-Box leverages advanced algorithms to generate real-time accompaniments based on user input. Musicians can enter chords and choose styles (and even let the computer generate these), and the software produces professional-quality backing tracks, making it a valuable tool for composers, arrangers, and performers seeking instant, high-quality music production (see Fig. 2).

Therefore, we can classify different music creations and choose more suitable algorithms to help with music creation. The following section introduces the related technologies and algorithms of AI music generation.

The processes of AI music generation

Similar to how the human brain creatively organizes information to create, AI-driven music production relies on training computers to analyze existing music and establish innovative rules for generating new compositions. This endeavor often entails feeding large datasets into computational systems for analysis. In the following sections, we delve into ML and explore notable algorithms associated with music creation.

ML techniques

ML can be categorized into two main types based on the underlying technology: supervised learning and unsupervised learning. Supervised learning involves training a model using known input–output pairs to make predictions about future outputs. On the other hand, unsupervised learning focuses on uncovering hidden patterns or inherent structures within input data (Mukhamediev, 2015).

Supervised learning

Supervised ML involves constructing a model capable of making predictions by leveraging available evidence amid uncertainty. Algorithms within supervised learning are fed with a set of known input data along with corresponding known output responses, enabling the model to generate accurate predictions for new data instances. This paradigm employs classification and regression techniques to facilitate the development of ML models. Classification models are adept at categorizing input data,

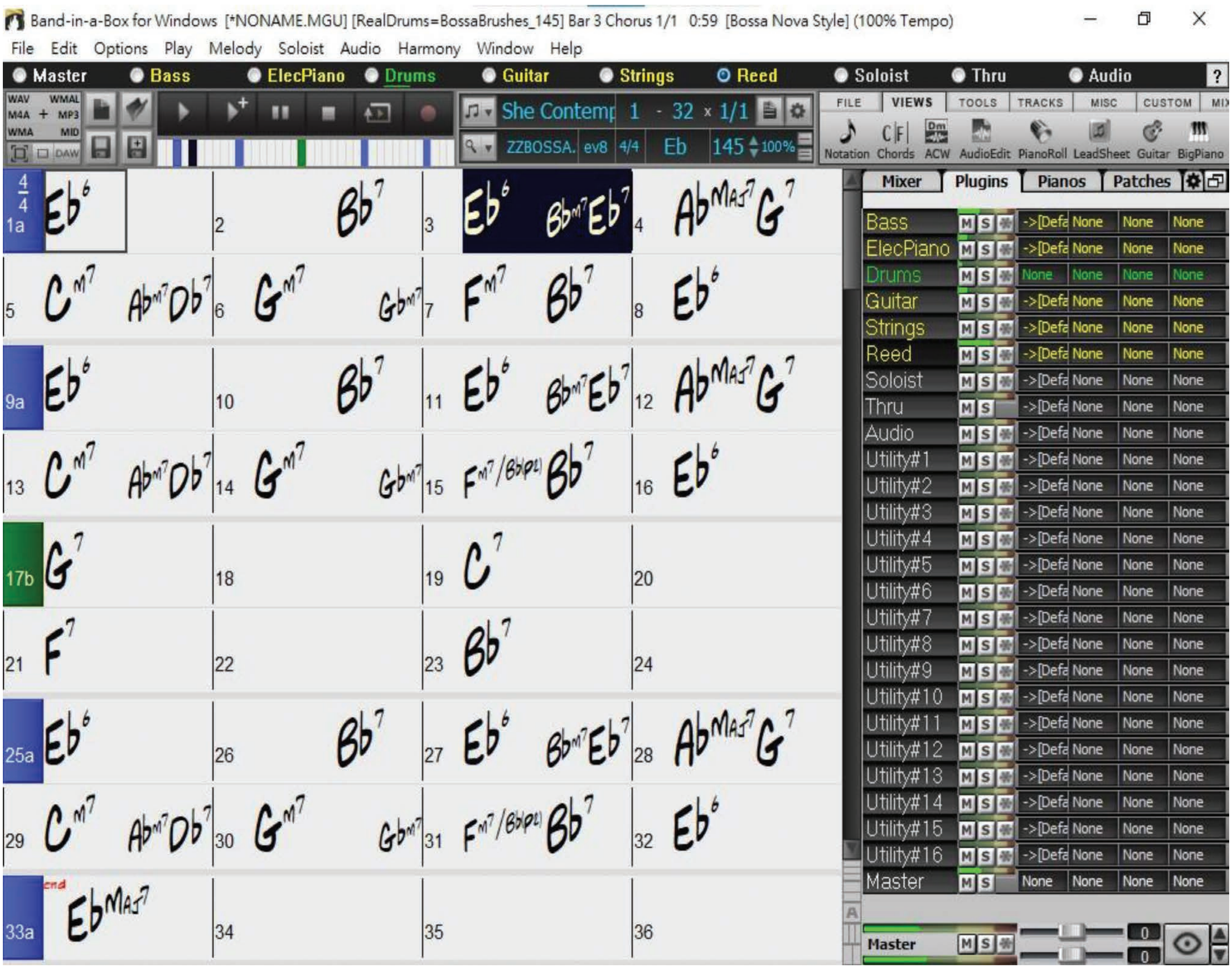

FIG2 A sample page screenshot of Band-in-a-Box computer-generated production.

finding utility in various domains, such as medical imaging, speech recognition, and credit scoring. Conversely, regression techniques are applied in scenarios like virtual sensing, power load forecasting, and algorithmic trading (MathWorks).

Unsupervised learning

Unsupervised learning represents a branch of ML characterized by its capacity to autonomously derive insights from data without human intervention. In contrast to supervised learning, where models are trained using labeled data, unsupervised ML operates on unlabeled datasets, enabling the algorithms to identify patterns and extract meaningful information without explicit guidance or instructions (Google Cloud Newsletter).

Examples of AI music composition algorithms

According to research by Siphocly et al. (2021), prominent AI music algorithms encompass various methodologies, each with distinctive features, as outlined here:

- *Rule-based reasoning*: Rule-based systems are adept at representing music theory and its underlying principles. Given that music theory inherently operates on rules, employing rule-based systems simplifies their integration into music composition processes.
- *Case-based reasoning (CBR)*: CBR systems excel in emulating human learning patterns by leveraging past musical experiences to inform future compositions.
- *Markov chains*: Well suited for melody generation, Markov chains excel at predicting subsequent notes based on prior musical knowledge.
- *Generative grammars*: Offering the swift creation of musical compositions in adherence to predefined musical rules, generative grammars facilitate efficient composition processes.
- *Linear programming*: Leveraging the concept of timbral space, linear programming has proven instrumental in synthesizing timbres, especially in early language descriptor applications.
- *Genetic algorithms (GAs)*: GAs have significantly contributed to various music creation tasks, including melody generation, timbre synthesis, and rhythm generation. Their resemblance to natural music creation processes enhances their efficacy in generating music.
- *Artificial immune systems (AISs)*: AISs stand out for providing multiple suggested solutions, particularly in chord progression generation.
- *Artificial neural networks (ANNs)*: ANNs, including recurrent neural networks and convolutional restricted Boltzmann machines, excel in capturing long-term dependencies in music composition. They generate music consistent with past patterns and possess the ability to produce polyphonic music and four-part choruses.
- *Deep neural networks (DNNs)*: DNNs produce music with considerable appeal to human listeners and are proficient at extracting musical features from training data without prior musical knowledge. DNNs generate works in a similar style to the training data, showcasing strong generalization capabilities. Additionally, convolutional neural networks provide a realistic simulation of the synthesis process by revisiting and enhancing previously generated melodic partials.
- *Generative adversarial networks (GANs)*: State-of-the-art GANs generate high-quality and engaging music by producing realistic compositions from random noise inputs.

The ongoing development of AI music creation involves optimizing algorithms for output works and refining and combining algorithms to enhance their complementary aspects, as noted by Carnovalini and Rodà (2020). There are currently a lot of AI-powered music software products on the market, each with different model combinations and design specifications, and they are usually constantly updated. Table 1 compares nine popular AI-powered music software programs currently available on the market: MuseNet, AIVA, Amadeus Code, Chrome Song Maker, Melobytes, Melody Sauce, Amper Music, Tracksy, and MusicGen. These examples of features can be used to evaluate various AIs, which is a general reference for music software. As a result, these comparisons provide an overview of the key features and capabilities of each AI-powered music software, helping users to choose the one that best fits their requirements and preferences.

Application of generative AI music

Generative AI tools influence music creation by providing unique ways to autonomously generate music. These tools can create melodies, harmonies, rhythms, and even entire songs based on predefined algorithms and training data. They enable music lovers to explore new creative avenues, experiment with unconventional structures, and enhance the compositional process. Generative AI can also help overcome compositional barriers, inspire new ideas, and open up endless possibilities for music production.

Use in movies, games, and advertising music

Music created by AI cannot yet replace the music tailor-made for high-standard movies, games, and advertisements by outstanding composers on the front line. However, to a certain extent, AI music can already meet the creation needs of certain image music. In movies, the goal of AI-driven music creation technology is to allow filmmakers to create customized soundscapes that evoke specific emotions, amplify tension, or emphasize key moments in a storyline. In addition, AI can help automate the scoring process, saving time and resources. In the film field, AI brands such as AIVA Technologies, Amper Music, and Mubert have made significant

TABLE 1. A comparison of nine popular AI-powered music software products currently available on the market.

AI-POWERED MUSIC SOFTWARE	FEATURES					
	ALGORITHM ADOPTION	USER INTERFACE	COMPATIBILITY WITH DAW OR OTHER RESOURCES	COLLABORATION AND SHARING CAPABILITIES	OUTPUT QUALITY	PRICING
MuseNet	Utilizes advanced AI algorithms for music generation across various genres	Simple interface with basic controls for generating music	Supports MIDI export for integration with DAWs	Limited collaboration features	Offers diverse and high-quality music generation capabilities	Registration required for access to the application programming interface Usage fees may apply
AIVA	Utilizes AI algorithms for composing classical music	Clean and intuitive interface	Provides MIDI export for integration with DAWs	Limited collaboration features	Known for producing realistic classical music compositions	Subscription-based pricing with different tiers based on usage
Amadeus Code	Uses AI algorithms for generating melodies and chord progressions	User-friendly interface with easy-to-use composition tools	Supports MIDI export for integration with DAWs	Limited collaboration features	Provides quality melodic ideas and chord progressions	Subscription-based pricing with various plans available
Chrome Song Maker	Utilizes simple AI algorithms for basic music composition	Web-based interface with intuitive controls	Limited integration options	Allows for easy sharing of compositions	Basic output suitable for casual music creation	Free to use
Melobytes	Uses AI algorithms for generating music based on images and text	Simple and straightforward interface	Limited integration options	Allows for sharing of compositions via URL	Varied depending on input More experimental in nature	Free to use with premium features available
Melody Sauce	Utilizes AI algorithms for generating melodic ideas and variations	User-friendly interface with intuitive controls	Supports MIDI export for integration with DAWs	Limited collaboration features	Provides diverse and customizable melodic ideas	One-time purchase with free updates
Amper Music	Utilizes AI algorithms for music composition and arrangement	Offers a user-friendly interface with intuitive controls	Compatible with major DAWs and supports MIDI export	Allows collaboration through cloud-based project sharing	Provides high-quality and customizable music compositions	Subscription-based pricing model with various plans available
Tracksy	Utilizes AI algorithms for generating music tracks across various genres	Simple interface with drag-and-drop functionality	Supports MIDI export and integration with DAWs	Limited collaboration features	Offers customizable music tracks with varying degrees of realism	Subscription-based pricing with different plans based on usage
MusicGen	Utilizes AI algorithms for generating music in specific styles and genres	Intuitive interface with preset options for quick music creation	Supports MIDI export for integration with DAWs	Limited collaboration features	Provides ready-made music suitable for background or production use	Subscription-based pricing with various plans available

DAW: digital audio workstation; MIDI: musical instrument digital interface.

progress in AI music creation. These companies use AI to generate music for movies, TV shows and other visual media, providing composers and filmmakers with innovative tools to enhance storytelling through music.

In the field of games, AI music creation plays an important role in adaptive and interactive soundtracks. These systems dynamically generate music based on player movements, in-game events, and the overall atmosphere, immersing players in a personalized listening experience. By incorporating AI algorithms that respond to game variables in real time, game developers can enhance engagement, increase replay value, and create more immersive gaming environments. In the gaming industry, AI-driven music creation is attracting attention from companies such as Hexany Audio, Endel, and Melodrive. These brands use AI algorithms to dynamically generate adaptive soundtracks for video games, creating an immersive audio experience that responds to player actions and enhances game engagement.

In the advertising world, by analyzing consumer behavior, preferences, and demographic data, AI can ideally produce music that fits the brand image, resonates with the target audience, and enhances the overall impact of the advertising campaign. AI-generated music may successfully evoke specific emotions, associations, and responses; enhance brand recognition; and cultivate an emotional connection with consumers. In the advertising space, brands such as Jukedeck, SKIO Music, Aflorithmic, and others are already leveraging AI music creation. These companies use AI technology to create customized music tracks for advertising campaigns, customizing sounds to match brand messages, target audience demographics, and emotional appeals, thereby increasing the effectiveness of marketing strategies through music.

The application of AI music creation in movies, games, and advertising opens up new possibilities to enhance storytelling, user engagement, and brand communication. The mentioned brands embody the innovative application of AI in music creation in the fields of film, games, and advertising, demonstrating the unique potential of technology in various interactive and narrative environments.

Applications in music education

In the world of learning and education, AI is revolutionizing traditional methods by providing innovative tools and platforms that enhance music teaching, practice, and creativity. AI applications in music learning can provide students with personalized feedback, instantly analyze their performance, and create tailored lesson plans to help them improve their skills effectively. These techniques cater to individual learning styles and rhythms, making music learning more interactive, engaging, and efficient (Lin and Yang, 2024). AI algorithms can assist music creation and help learners try different music structures, styles, and genres. By providing suggestions, generating accompaniments, and even composing melodies, AI tools enable students to explore their creativity and develop their musical voices.

Furthermore, the application of AI in music education bridges the gap between theory and practice. The AI-powered platform can simulate ensemble performance, provide virtual rehearsal spaces, and provide collaborative opportunities for students to interact with peers and musicians around the world, cultivating an immersive, collaborative learning experience. Overall, the application of AI in music learning and education is changing the way individuals engage with music, providing new avenues for exploration, skill development, and creative expression in technologically advanced and interconnected music environments.

Innovative practice in music performance

By using AI algorithms and ML technology to develop tools and systems to assist musicians in improvisation, composition, and interpretation, new and innovative ways of music performance can be brought about, such as performances by ensembles of musicians and "AI virtual musicians" (Su, 2021). In addition, AI can also create an immersive performance experience for musicians and audiences by connecting 3D objects, 3D scenes, volumetric photography, special effects, and other interactive systems and technologies, transforming traditional concerts into multisensory and interactive activities that enable musicians to push the boundaries of traditional performance practice.

Social, political, and ethical implications

Originality and copyright issues

AI music creation raises significant questions about the originality and copyright of the resulting works. Because AI systems can analyze large amounts of existing music data and generate compositions based on learned patterns, there is a potential challenge in determining the true author of AI-generated music. This blurring of authorship raises concerns about intellectual property rights and copyright ownership, as traditional legal frameworks may struggle to address the unique nature of AI-created content. After AI creation has become a trend, people will need to reevaluate copyright law and the protection of creative works.

Legal scholars have proposed that user works can be regarded as joint works, and joint authors include the user of the app and the AI programmer or trainer or his employer (Chen, 2023). In addition, regarding the issue of the ownership of copyrights, it may be necessary to amend the law to create copyright owners or legal users of "musical works generated by computer programs" to deal with inevitable issues of ownership of rights and responsibilities in the future.

Impact on music professionals

AI technology-assisted creation has both a transformative and disruptive impact on music professionals. AI tools can increase the productivity of composers, producers, and artists; inspire creativity; and streamline the music production process. However, this is an optimistic observation without considering market supply and demand issues. Additionally, AI's automation of certain tasks, such as melody creation and mastering, may reduce the need for traditional roles in the music industry. In scenes that have low requirements for work quality or personalization, such as the background music played in some restaurants, there are gradually many cases of "employing" AI music creation.

Today, available jobs for musicians are disappearing, and the conditions for making a living through creation will become increasingly stringent in the future. Most of these music jobs, which have been gradually replaced by AI, were an option for novice creatives or students at the entry-level stage, who could learn and earn living expenses at the same time. Therefore, the large-scale development of AI music may make it more difficult for industry entry-level talents to generate income from music. Not only that, using AI to copy established music styles will significantly lower the threshold for creation. A large amount of music of the same type may weaken the artistic diversity and originality of the music industry. As a result, music professionals are challenged to adapt to AI-driven workflows, collaborate with intelligent systems, and leverage AI tools to remain competitive in a rapidly evolving industry environment.

Growth and development of music learners

AI in music education provides opportunities for personalized learning experiences, skill development, and musical exploration. The AI-driven platform can provide tailored feedback, practice suggestions, and interactive learning modules to enhance learning and make learning more enjoyable. Taking the interactive electric guitar teaching series published by Kamo and Matsushita in 2022 as an example, learners can intuitively obtain action feedback on the playing status. AI can also provide virtual practice partners and personalized learning courses. AI tools lower the threshold for creative learning, and combining them with technological tools can increase the fun of learning. The application of AI can encourage more people to invest in creation, help generate new aesthetic experiences, and make it easier for learners to review their own learning and correct mistakes. As far as education itself is concerned, AI brings vast possibilities and opportunities to music learning and creation.

Cultural diversity and the challenges of AI music generation

The reliance of AI systems on existing datasets and algorithms may bias creation toward mainstream genres, homogenize musical expression, and limit the expression of different cultural traditions and nonmainstream—but very unique—artistry. Promoting inclusivity and cultural awareness as well as reflecting the richness and diversity of global musical traditions in AI music creation will require continued work.

Outlook

The collaboration between humans and AI in music creation fuses human creativity, emotion, and intuition with AI's computing power, analytical capabilities, and creative algorithms. This partnership transcends traditional notions of authorship, resulting in hybrid musical compositions that blend human and machine. In this symbiotic relationship, AI serves as a cocreator, source of inspiration, and catalyst for experimentation, challenging artists to push the boundaries of musical expression and explore new realms of sound. We can expect advancements in AI's ability to generate highly sophisticated and emotionally resonant music, expanding beyond current genres to create entirely new forms of musical expression. The development of more intuitive and interactive AI tools will empower musicians to collaborate with AI in real time, leading to unprecedented creative synergies.

However, these advancements come with significant challenges. One primary concern is the ethical and legal implications surrounding authorship and copyright. As AI systems become more autonomous, determining the actual creator of a piece of music becomes increasingly complex, necessitating new legal frameworks to address ownership and intellectual property rights. Additionally, there is the risk of homogenization in musical output, as AI-generated music might lean heavily on existing popular styles, potentially stifling diversity and innovation in the music industry. Moreover, the widespread adoption of AI in music production could disrupt traditional roles and job opportunities for musicians, especially those at the entry level. As AI takes over tasks such as composition and production, music professionals must adapt by developing new skills and collaborating effectively with these technologies.

Despite these challenges, the collaboration between humans and AI in music creation heralds a new era of creative possibility. Boundaries will continue to blur conventions will evolve, and the art of music will be redefined, fostering a culture of experimentation, innovation, and interdisciplinary collaboration. By embracing these changes with an open mind, musicians and technologists alike can unlock new opportunities for artistic growth and innovation, shaping the future of music in exciting and unforeseen ways.

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Read more about it

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About the authors

Tsen-Fang Lin (tflin@stust.edu.tw) earned her master’s degree in composition from the Department of Music, National Sun Yat-sen University, Kaohsiung, Taiwan, and her Ph.D. degree in electronic engineering from the Southern Taiwan University of Science and Technology, Tainan, Taiwan. She is an assistant professor in the Department of Popular Music Industry, Southern Taiwan University of Science and Technology, Tainan 710301, Taiwan. She has created many computer and electronic music compositions and obtained a number of patents related to sonic applications.

Liang-Bi Chen (liangbi.chen@gmail.com) earned his Ph.D. degree in electronic engineering from Southern Taiwan University of Science and Technology, Tainan, Taiwan. Currently, he is an associate professor with and chair of the Department of Computer Science and Information Engineering, National Penghu University of Science and Technology, Penghu 880011, Taiwan. He serves as an associate editor for *IEEE Access* and *IEEE Transactions on Consumer Electronics* and as the chair of the Internet of Things Technical Committee of the IEEE Consumer Technology Society. He is a Senior Member of IEEE.